

Bank Transaction Database (MongoDB)

Name: Sagi Sahasra Srihasa

Roll: 2520030008

1)Create database and add collections

The screenshot shows the MongoDB Data Explorer interface. On the left, the 'CLUSTERS (1)' sidebar shows a tree structure: Cluster0 > BankDB > customers (highlighted with a blue border). The main panel shows the 'Cluster0 > BankDB > customers' view. The 'Documents' tab is active, showing 0 documents. Below the tabs, there is a search bar with the placeholder text 'Type a query: { field: 'value' } or [Generate query](#)'. Below the search bar are three buttons: 'ADD DATA', 'BULK', and 'EXPORT CODE'.

2)Insert data in customers:

The screenshot shows the MongoDB Data Explorer interface with data inserted into the 'customers' collection. The 'Documents' tab is active, showing 8 documents. The first document is displayed in a code block:

```
_id: ObjectId('6a891b2f8c3a88119bd4c8ef')
customer_id: 101
name: "Ravi Kumar"
city: "Hyderabad"
account_type: "Savings"
balance: 85000
status: "Active"
```

The second document is also displayed in a code block:

```
_id: ObjectId('6a891b3b8c3a88119bd4c8f1')
customer_id: 102
name: "Priya Sharma"
city: "Vijayawada"
account_type: "Current"
balance: 125000
status: "Active"
```

At the bottom right, there is a pagination control showing '25' and '1 - 8 of 8'.

ORGANIZATION
Sahasra

PROJECT
Books_

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

BankDB

customers

transactions

admin

bookflow

local

Welcome

customers

transactions

+

Cluster0 > BankDB > customers

Documents 8 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

status : "Active"

_id: ObjectId('6a891b4c8c3a88119bd4c8f3')

customer_id : 103

name : "Arjun Reddy"

city : "Hyderabad"

account_type : "Savings"

balance : 45000

status : "Active"

_id: ObjectId('6a891b568c3a88119bd4c8f5')

customer_id : 104

name : "Sneha Rao"

city : "Warangal"

ORGANIZATION
Sahasra

PROJECT
Books_

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

BankDB

customers

transactions

admin

bookflow

local

Welcome

customers

transactions

+

Cluster0 > BankDB > customers

Documents 8 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

_id: ObjectId('6a891b618c3a88119bd4c8f7')

customer_id : 105

name : "Kiran Kumar"

city : "Guntur"

account_type : "Current"

balance : 65000

status : "Inactive"

_id: ObjectId('6a891b6b8c3a88119bd4c8f9')

customer_id : 106

name : "Anjali Singh"

city : "Hyderabad"

account_type : "Savings"

balance : 175000

status : "Active"

ORGANIZATION
Sahasra

PROJECT
Books_

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDB
 - customers
 - transactions
- admin
- bookflow
- local

Welcome

customers

transactions

+

Cluster0 > BankDB > customers

Documents (8) Aggregations Schema Indexes (1) Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

```
customer_id : 107
name : "Rahul Verma"
city : "Bengaluru"
account_type : "Current"
balance : 95000
status : "Active"
```

```
_id : ObjectId('6a891bb98c3a88119bd4c8fd')
customer_id : 108
name : "Meena Devi"
city : "Chennai"
account_type : "Savings"
balance : 320000
status : "Active"
```

3) Insert data in transactions

ORGANIZATION
Sahasra

PROJECT
Books_

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDB
 - customers
 - transactions
- admin
- bookflow
- local

Welcome

customers

transactions

+

Cluster0 > BankDB > transactions

Documents (15) Aggregations Schema Indexes (1) Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

```
_id : ObjectId('6a891c118c3a88119bd4c900')
transaction_id : 1001
customer_id : 101
type : "Deposit"
amount : 25000
mode : "UPI"
date : "2026-08-01"
status : "Success"
```

```
_id : ObjectId('6a891c1b8c3a88119bd4c902')
transaction_id : 1002
customer_id : 101
type : "Withdrawal"
amount : 10000
mode : "ATM"
```

ORGANIZATION
Sahasra

PROJECT
Books_

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDB
 - customers
 - transactions
- admin
- bookflow
- local

Welcome

customers

transactions

+

Cluster0 > BankDB > transactions

Documents 15AggregationsSchemaIndexes 1Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

```
_id: ObjectId('6a891c2d8c3a88119bd4c904')
transaction_id: 1003
customer_id: 102
type: "Deposit"
amount: 50000
mode: "NEFT"
date: "2026-08-02"
status: "Success"
```

```
_id: ObjectId('6a891c378c3a88119bd4c906')
transaction_id: 1004
customer_id: 102
type: "Withdrawal"
amount: 15000
mode: "ATM"
```

ORGANIZATION
Sahasra

PROJECT
Books_

Co

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDB
 - customers
 - transactions
- admin
- bookflow
- local

Welcome

customers

transactions

+

Cluster0 > BankDB > transactions

Documents 15AggregationsSchemaIndexes 1Validation

Type a query: { field: 'value' } or [Generate query](#) [Explain](#)

ADD DATA

BULK

EXPORT CODE

251 - 15 of 15

```
_id: ObjectId('6a891c448c3a88119bd4c905')
transaction_id: 1005
customer_id: 103
type: "Deposit"
amount: 30000
mode: "UPI"
date: "2026-08-03"
status: "Success"
```

```
_id: ObjectId('6a891c4f8c3a88119bd4c90a')
transaction_id: 1006
customer_id: 103
type: "Withdrawal"
amount: 5000
mode: "ATM"
date: "2026-08-04"
```

ORGANIZATION
Sahasra

PROJECT
Books_

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDB
 - customers
 - transactions
- admin
- bookflow
- local

Welcome

customers

transactions

+

Cluster0 > BankDB > transactions

Documents 15AggregationsSchemaIndexes 1Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

25

```
transaction_id: 1007
customer_id: 104
type: "Deposit"
amount: 100000
mode: "NEFT"
date: "2026-08-04"
status: "Success"
```

```
_id: ObjectId('6a891c648c3a88119bd4c90e')
transaction_id: 1008
customer_id: 104
type: "Withdrawal"
amount: 25000
mode: "ATM"
date: "2026-08-05"
status: "Success"
```

ORGANIZATION
Sahasra

PROJECT
Books_

Compare

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDB
 - customers
 - transactions
- admin
- bookflow
- local

Welcome

customers

transactions

+

Cluster0 > BankDB > transactions

Documents 15AggregationsSchemaIndexes 1Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

25

1 - 15 of 15

```
customer_id: 105
type: "Deposit"
amount: 15000
mode: "UPI"
date: "2026-08-05"
status: "Success"
```

```
_id: ObjectId('6a891c768c3a88119bd4c912')
transaction_id: 1010
customer_id: 106
type: "Deposit"
amount: 75000
mode: "NEFT"
date: "2026-08-06"
status: "Success"
```

View monitor

Explain

ORGANIZATION
Sahasra

PROJECT
Books_

Con

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDB
 - customers
 - transactions
- admin
- bookflow
- local

Welcomecustomerstransactions+

Cluster0 > BankDB > transactions

View

Documents15AggregationsSchemaIndexes1Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

ADD DATA

BULK

EXPORT CODE

251 - 15 of 15

```
customer_id : 100
type : "Withdrawal"
amount : 20000
mode : "ATM"
date : "2026-08-06"
status : "Success"

_id: ObjectId('6a891c8d8c3a88119bd4c916')
transaction_id : 1012
customer_id : 107
type : "Deposit"
amount : 40000
mode : "IMPS"
date : "2026-08-07"
status : "Success"
```

ORGANIZATION
Sahasra

PROJECT
Books_

Compare tiers

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDB
 - customers
 - transactions
- admin
- bookflow
- local

Welcomecustomerstransactions+

Cluster0 > BankDB > transactions

View monitoring

Documents15AggregationsSchemaIndexes1Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

ADD DATA

BULK

EXPORT CODE

251 - 15 of 15

```
type : "Withdrawal"
amount : 12000
mode : "ATM"
date : "2026-08-07"
status : "Success"

_id: ObjectId('6a891ca08c3a88119bd4c91a')
transaction_id : 1014
customer_id : 108
type : "Deposit"
amount : 150000
mode : "NEFT"
date : "2026-08-08"
status : "Success"
```

ORGANIZATION Sahasra > PROJECT Books_ > Compare tiers

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - BankDB
 - customers
 - transactions**
 - admin
 - bookflow
 - local

Welcome customers transactions +

Cluster0 > BankDB > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN RES

ADD DATA BULK EXPORT CODE

25 1 - 15 of 15

```
type : "Deposit"
amount : 150000
mode : "NEFT"
date : "2026-08-08"
status : "Success"

_id: ObjectId('6a891ca98c3a88119bd4c91c')
transaction_id : 1015
customer_id : 108
type : "Withdrawal"
amount : 30000
mode : "ATM"
date : "2026-08-08"
status : "Success"
```

4) Display all customers;

ORGANIZATION Sahasra > PROJECT Books_ > Compare tiers

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - BankDB
 - customers**
 - transactions
 - admin
 - bookflow
 - local

Welcome customers transactions +

Cluster0 > BankDB > customers

Documents 8 Aggregations Schema Indexes 1 Validation

Generate

ADD DATA BULK EXPORT CODE

customers

	_id ObjectId	customer_id Int32	name String
1	ObjectId('6a891b2f8c3a881...	101	"Ravi Kumar"
2	ObjectId('6a891b3b8c3a881...	102	"Priya Sharma"
3	ObjectId('6a891b4c8c3a881...	103	"Arjun Reddy"
4	ObjectId('6a891b568c3a881...	104	"Sneha Rao"
5	ObjectId('6a891b618c3a881...	105	"Kiran Kumar"
6	ObjectId('6a891b6b8c3a881...	106	"Anjali Singh"
7	ObjectId('6a891b758c3a881...	107	"Rahul Verma"
8	ObjectId('6a891bb98c3a881...	108	"Meena Devi"

5) Find active customers:

ORGANIZATION Sahasra > PROJECT Books_

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

BankDB

customers

transactions

admin

bookflow

local

Welcome | customers | transactions | +

Cluster0 > BankDB > customers

Documents 8 | Aggregations | Schema | Indexes 1 | Validation

Generate query

EXPLAIN

ADD DATA | BULK | EXPORT CODE

25 | 1 - 7 of 7

_id	ObjectId	customer_id	name	city
1	ObjectId('6a891b2f8c3a881...')	101	"Ravi Kumar"	"Hyderabad"
2	ObjectId('6a891b3b8c3a881...')	102	"Priya Sharma"	"Vijayawada"
3	ObjectId('6a891b4c8c3a881...')	103	"Arjun Reddy"	"Hyderabad"
4	ObjectId('6a891b568c3a881...')	104	"Sneha Rao"	"Warangal"
5	ObjectId('6a891b6b8c3a881...')	106	"Anjali Singh"	"Hyderabad"
6	ObjectId('6a891b758c3a881...')	107	"Rahul Verma"	"Bengaluru"

6) Find customers with balance greater than 100000:

ORGANIZATION Sahasra > PROJECT Books_

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

BankDB

customers

transactions

admin

bookflow

local

Welcome | customers | transactions | +

Cluster0 > BankDB > customers

Documents 8 | Aggregations | Schema | Indexes 1 | Validation

Generate query

EXPLAIN

ADD DATA | BULK | EXPORT CODE

25 | 1 - 7 of 7

_id	ObjectId	customer_id	name
1	ObjectId('6a891b3b8c3a881...')	102	"Priya Sharma"
2	ObjectId('6a891b568c3a881...')	104	"Sneha Rao"
3	ObjectId('6a891b6b8c3a881...')	106	"Anjali Singh"
4	ObjectId('6a891bb98c3a881...')	108	"Meena Devi"

7) Find Hyderabad customers:

ORGANIZATION Sahasra > PROJECT Books_

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

BankDB

customers

transactions

admin

bookflow

local

Welcome customers transactions +

Cluster0 > BankDB > customers

Documents 8 Aggregations Schema Indexes 1 Validation

{ "city": "Hyderabad" }

Generate query

ADD DATA BULK EXPORT CODE

25 1 - 3

	_id ObjectId	customer_id Int32	name String	city
1	ObjectId('6a891b2f8c3a881...	101	"Ravi Kumar"	"Hyderabad"
2	ObjectId('6a891b4c8c3a881...	103	"Arjun Reddy"	"Hyderabad"
3	ObjectId('6a891b6b8c3a881...	106	"Anjali Singh"	"Hyderabad"

8) Find savings account customers

ORGANIZATION Sahasra > PROJECT Books_

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

BankDB

customers

transactions

admin

bookflow

local

Welcome customers transactions +

Cluster0 > BankDB > customers

Documents 8 Aggregations Schema Indexes 1 Validation

{ "account_type": "Savings" }

Generate query

ADD DATA BULK EXPORT CODE

25

	_id ObjectId	customer_id Int32	name String
1	ObjectId('6a891b2f8c3a881...	101	"Ravi Kumar"
2	ObjectId('6a891b4c8c3a881...	103	"Arjun Reddy"
3	ObjectId('6a891b568c3a881...	104	"Sneha Rao"
4	ObjectId('6a891b6b8c3a881...	106	"Anjali Singh"
5	ObjectId('6a891bb98c3a881...	108	"Meena Devi"

9) Find ATM transactions

ORGANIZATION Sahasra > PROJECT Books_ > Complete

Data Explorer

{ } My Queries
⚙ Data Modeling

CLUSTERS (1)
Search clusters
Cluster0
BankDB
customers
transactions

Welcome | customers | transactions +

Cluster0 > BankDB > transactions View more

Documents 15 Aggregations Schema Indexes 1 Validation

```
{  
  "mode": "ATM"  
}
```

[Generate query](#) [Explain](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#) 25 1 - 7 of 7

	_id ObjectId	transaction_id Int32	customer_id Int32	type String
1	ObjectId('6a891c1b8c3a881...')	1002	101	"Withdrawal"
2	ObjectId('6a891c378c3a881...')	1004	102	"Withdrawal"
3	ObjectId('6a891c4f8c3a881...')	1006	103	"Withdrawal"
4	ObjectId('6a891c648c3a881...')	1008	104	"Withdrawal"
5	ObjectId('6a891c808c3a881...')	1011	106	"Withdrawal"
6	ObjectId('6a891c978c3a881...')	1013	107	"Withdrawal"

10) FIND DEPOSITS GREATER THAN 50000

ORGANIZATION Sahasra > PROJECT Books_ > Complete

Data Explorer

{ } My Queries
⚙ Data Modeling

CLUSTERS (1)
Search clusters
Cluster0
BankDB
customers
transactions

Welcome | customers | transactions +

Cluster0 > BankDB > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

```
{  
  "type": "Deposit",  
  "amount": {  
    "$gt": 50000  
  }  
}
```

[Generate query](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#) 25

	_id ObjectId	transaction_id Int32	customer_id Int32
1	ObjectId('6a891c5a8c3a881...')	1007	104
2	ObjectId('6a891c768c3a881...')	1010	106
3	ObjectId('6a891ca08c3a881...')	1014	108

11) FIND HYDERABAD CUSTOMERS WITH HIGH BALANCE

ORGANIZATION Sahasra > PROJECT Books_

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

BankDB

customers

transactions

admin

bookflow

local

Welcome | customers | transactions +

Cluster0 > BankDB > customers

Documents 8 | Aggregations | Schema | Indexes 1 | Validation

Generate query

```
{
  "city": "Hyderabad",
  "balance": {
    "$gt": 100000
  }
}
```

ADD DATA | BULK | EXPORT CODE

25

_id	customer_id	name
ObjectId('6a891b6b8c3a881...	106	"Anjali Singh"

12) SORT CUSTOMERS BY BALANCE

ORGANIZATION Sahasra > PROJECT Books_ Compare

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

BankDB

customers

transactions

admin

bookflow

local

Welcome | customers | transactions +

Cluster0 > BankDB > customers View monitor

Documents 8 | Aggregations | Schema | Indexes 1 | Validation

\$sort

Generate aggregation

Export

Untitled - modified | SAVE | CREATE NEW | EXPORT CODE | PREVIEW | STAGES

Stage 1 \$sort

```
1 {
2   "balance": -1
3 }
4
```

Output preview after \$sort stage (Sample of 8 document)

```
_id: ObjectId('6a891bb98c3a88119bd4c8fd')
customer_id: 108
name: "Meena Devi"
city: "Chennai"
account_type: "Savings"
balance: 320000
status: "Active"
```

13) CALCULATE TOTAL TRANSACTION AMOUNT

ORGANIZATION
Sahasra

PROJECT
Books_

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDB
 - customers
 - transactions
- admin
- bookflow
- local

Welcome

customers

transactions

+

Cluster0 > BankDB > transactions

Documents 15

Aggregations

Schema

Indexes 1

Validation

Sort

\$group

Generate aggregation

pi - modified

SAVE

+ CREATE NEW

EXPORT CODE

PREVIEW

Stage 2 \$group

```
1 {
2   "_id": null,
3   "total_amount": {
4     "$sum": "$amount"
5   }
6 }
```

Output preview after \$group stage (San

```
_id: null
total_amount : 602000
```